

REVISED PROJECT PLAN: WEEKS 11-16

OffGrid English - Enhanced Implementation Plan

EXECUTIVE SUMMARY

Total Features: 21 (11 from your suggestions + 10 from recommendations) **Timeline:** 6 weeks (Weeks 11-16) **Cost:** \$0 (all free, open-source tools) **Offline-First:** Maintained throughout **Research-Grounded:** All features map to Assignments 1-3 + Qualifier Question

FEATURE INVENTORY (21 TOTAL)

Tier 1: CORE Features (10) - Highest Priority

#	Feature	Research Source	Week
1	Contrastive Analysis View	A1: #5,#8; A3: Sokeng 2014	12
2	Personalized Learning Path	A2: SLA; A3: #24; QQ	12
3	Spaced Repetition System	A2: Krashen; A3: persistence	12
5	Real-World Cameroonian Scenarios	A1: #2; A3: authentic tasks	11
7	Enhanced Teacher Validation	QQ: sampling; Proposal	11
9	False Cognates Highlighter	A1: #8; A2: transfer	15
10	Teacher Q&A Queue	A1: #7 (WhatsApp COVID)	15
12	Voice Input for Answers	A1: #9,#10,#13; A3: #7	13
19	Enhanced Analytics Dashboard (charts)	A3: #17; QQ: metrics	11
-	CSV Exports (all data)	QQ: open standards	11

Tier 2: STRONG Features (5) - High Priority

#	Feature	Research Source	Week
6	Interactive Error Explanations	A2: explicit; A3: #15,#16	11
8	Achievement System	A1: #18; A3: #12	12
11	Enhanced Progress Tracking	A3: persist; QQ: metrics	12
21	Bookmarking	A3: #14,#24	12

Tier 3: SUPPORTED Features (6) - Medium Priority

#	Feature	Research Source	Week
4	Weekly Progress Report (PNG/PDF)	A3: metrics; QQ: equity	13
13	Dark Mode	A2: UNESCO; accessibility	11

#	Feature	Research Source	Week
14	Micro-interactions & Animations	A3: #17 (cognitive load)	11-13
15	Mobile-First Optimizations	A1: #11,#12,#13 (MALL)	13
16	PWA Excellence	A3: #19; Proposal	13
17	Professional Onboarding	A2: #13; A3: #10	13
18	Challenge Mode	A1: #18; A3: #12	12
20	Certificate Generator	A3: #24 (motivation)	16

WEEK-BY-WEEK BREAKDOWN

WEEK 11: Enhanced UI/UX + Analytics Foundation (Nov 4-10)

Theme: Transform from functional to professional + Data visualization

Goals: - Complete teacher validation interface enhancement - Build visual analytics dashboard with charts
- Add dark mode - Integrate Cameroonian scenarios - Setup animation framework

Tasks:

Task	Hours	Definition of Done
Install Dependencies	1	<code>npm install framer-motion recharts</code> succeeds; TypeScript types work
Enhanced Analytics Dashboard	8	- 5 metric categories render as tables- Recharts installed and working- 3 charts: completion bar, accuracy line, transfer comparison- All charts responsive on mobile- CSV export includes APP_VERSION
Enhanced Teacher Validation	4	- Keyboard shortcuts (A=agree, D=disagree, ←/→ navigate)- Split-screen view (student vs correct answer)- Sample counter shows progress (X of 20)- Agreement rate updates in real-time- validation.csv exports correctly
Dark Mode	3	- System preference detection (<code>prefers-color-scheme</code>)- Manual toggle in settings- Preference saved to localStorage- All colors have dark variants- WCAG 4.5:1 contrast maintained
Real-World Cameroonian Scenarios	5	- Add <code>scenario</code> field to item schema- 3-5 scenarios implemented (market, moto-taxi, school, job interview)- Each scenario has 4-6 items- Simple icon/emoji per scenario- Contextual feedback references scenario
Interactive Error Explanations (setup)	3	- Framer Motion basic animations working- Step-by-step feedback component created- Test with 2-3 sample explanations- Animations smooth on low-end devices
Micro-interactions Foundation	2	- Button ripple effects on tap- Success animations (checkmark fade-in)- Loading skeleton screens- Smooth page transitions
Code Review & Testing	2	- All TypeScript strict mode passes- ESLint clean- Manual testing on Android emulator- Lighthouse accessibility 90

Total: 28 hours **Deliverable:** Polished dashboard with visual analytics, enhanced teacher tools, dark mode working

WEEK 12: Smart Learning Features (Nov 11-17)

Theme: Intelligence layer - Adaptive learning + Spaced repetition

Goals: - Implement personalized learning path algorithm - Add spaced repetition system - Build achievement system - Enhanced progress tracking - Bookmarking feature

Tasks:

Task	Hours	Definition of Done
Contrastive Analysis View	6	- Toggle button on each question: “ French Comparison”- Shows side-by-side French/English structures- Visual highlighting (red=interference, green=correct)- Explains “Why French speakers struggle”- Data stored in item schema <code>frenchComparison</code> field- Add to 20-30 high-frequency error items
Personalized Learning Path	8	- Algorithm scores weak areas (by module accuracy)- “Smart Practice” mode button- Selects 10 questions from weakest 2 categories- Visual skill tree shows mastery % per module- Algorithm uses IndexedDB queries (no external API)- Works offline
Spaced Repetition System	6	- Track <code>lastAttempted</code> timestamp per item- Track <code>difficultyScore</code> (0-1, based on attempts)- “Review Mode” surfaces items: wrong + not seen in 3+ days- SM-2 algorithm (simplified): intervals 1d, 3d, 7d, 14d- Visual indicator: “Due for review” badge- Stores in IndexedDB
Achievement System	5	- 5 achievement types: 1. First Module Complete 2. Perfect Score (10/10) 3. 5-Day Streak 4. 100 Questions Answered 5. All Modules Complete- Toast notification on unlock- Achievements page lists all (locked/unlocked)- Stored in localStorage (offline)
Enhanced Progress Tracking	4	- Visual progress rings (circular, animated)- Module completion badges- Unlock system: Form B locked until Form A 70%- Overall progress dashboard- Works offline with IndexedDB
Bookmarking	3	- “ Bookmark” button on question screen- “My Bookmarks” page lists all saved items- Shows module, question text, difficulty- Remove bookmark option- Stored in IndexedDB
Challenge Mode	4	- “Speed Challenge”: 10 questions, 2-min timer- “Accuracy Challenge”: Get 10 correct in a row- “Streak Challenge”: Practice 7 days straight- Results screen shows best scores- Leaderboard (local only, no sync)
Testing & Polish	2	- All features tested on emulator- TypeScript strict mode passes- Performance check (no lag on interactions)

Total: 38 hours **Deliverable:** Intelligent learning system with adaptive paths and spaced repetition

WEEK 13: Mobile Excellence + Milestone 2 (Nov 18-24)

Theme: Perfect mobile experience + Second milestone

Goals: - Mobile-first optimizations - PWA excellence - Voice input - Weekly progress reports - Professional onboarding - Milestone 2 video

Tasks:

Task	Hours	Definition of Done
Mobile-First Optimizations	5	- Touch targets 48px (WCAG compliant)- Swipe gestures: left/right for next/prev question- Bottom navigation for thumb zone- Pull-to-refresh on lists- Test on 3 different screen sizes
PWA Excellence	4	- Custom splash screen (1080x1920, logo + branding)- App icons: 192x192, 512x512, maskable- Native feel: no browser chrome, status bar color- Install prompt on first visit- Smooth animations (60fps)- Tests: PWABuilder passes all checks
Voice Input for Answers	6	- Web Speech API integration (<code>webkitSpeechRecognition</code>)- Microphone permission request- “ Speak Answer” button (optional)- Fallback: Click to type if denied/unsupported- Visual feedback (listening animation)- Validates spoken answer against correct answer- Logs pronunciation attempts (for teacher review)- Graceful degradation on unsupported browsers
Weekly Progress Report	5	- HTML Canvas generates PNG summary- Shows: streak, modules completed, accuracy trend, top achievement- Cameroonian color accents (green/yellow/red)- “Share” button uses Web Share API- Fallback: Download PNG if share unsupported- “Export PDF” option (browser print to PDF)- Triggered automatically every 7 days
Professional Onboarding	4	- 4-screen welcome flow on first launch: 1. Welcome + app purpose 2. How it works (3-phase learning) 3. Offline demo (show airplane mode icon) 4. Choose your first module- Swipeable cards (react-swipeable)- “Skip” option- Sets <code>onboardingCompleted</code> flag in <code>localStorage</code>
Performance Optimization	3	- Lazy load routes: <code>React.lazy()</code> for Analytics, Validation- Image optimization: Use WebP with PNG fallback- Virtual scrolling for long module lists- Bundle analysis: Keep <250KB gzipped- Code splitting by route
Milestone 2 Video Preparation	5	- Record 5-min demo video: 1. Show all 7 modules 2. Demonstrate 5 new features (contrastive analysis, spaced repetition, voice input, analytics charts, achievements) 3. Show offline functionality proof 4. CSV export demonstration 5. Mobile responsiveness- Edit with screen recording + voiceover- Upload to required platform

Total: 32 hours **Deliverable:** App-like mobile experience + Milestone 2 submission

WEEK 14: Testing & Quality Assurance (Nov 25-Dec 1)

Theme: Ensure quality, reliability, and accessibility

Goals: - Comprehensive test suite - Real device testing - Bug fixes - Accessibility audit

Tasks:

Task	Hours	Definition of Done
Playwright E2E Test Suite	6	- Test critical flows: 1. Module selection → practice → feedback 2. Offline mode: complete item, check IndexedDB 3. Teacher validation: sample review, CSV export 4. Analytics: view metrics, export CSV 5. Voice input: permission handling- Automated zero-byte test passes- CI integration (runs on push)- Screenshots on failure
Lighthouse CI	2	- Performance 90- Accessibility 90- Best Practices 90- PWA 90- SEO 80- CI integration
Accessibility Audit (axe-core)	3	- Install @axe-core/react - Run automated scan- Fix critical issues: - Missing ARIA labels - Contrast violations - Focus indicators - Keyboard navigation- Manual screen reader test (VoiceOver/TalkBack)
Real Device Testing	4	- Test on 3 physical devices: 1. Android (mid-range, Chrome) 2. Android (low-end, < 2GB RAM) 3. iOS (Safari)- Test scenarios: - Offline mode works - Voice input works (or fails gracefully) - Swipe gestures smooth - Animations not janky - Storage quota doesn't exceed- Document device compatibility
Visual Regression Testing	2	- Install playwright-test with screenshots- Baseline screenshots for: - Home page - Module learning page - Practice page - Analytics dashboard - Dark mode variants- Compare against baseline on changes
Bug Fixes	6	- Fix all P0/P1 bugs from testing- Address accessibility violations- Handle edge cases: - Storage quota exceeded - Network interruption mid-sync - Multiple tabs open - Resume from where left off- Data integrity checks
Performance Testing	2	- Test on slow 3G network (Chrome DevTools)- Ensure offline mode works with 0 network- Check bundle size: 250KB gzipped- Test with 1000+ attempts in IndexedDB- Memory leak check (Chrome DevTools)
Code Quality Review	3	- ESLint: 0 warnings- TypeScript: strict mode, no any- Code coverage 70% (critical paths)- Remove console.logs (production build)- Add JSDoc comments to complex functions

Total: 28 hours **Deliverable:** Comprehensive test suite, bug-free experience, accessibility compliant

WEEK 15: Evaluation & Advanced Features (Dec 2-8)

Theme: User testing, expert review, final features

Goals: - User testing with 3-5 students - Expert review with ESL educators - False cognates feature - Teacher Q&A queue - Advanced analytics

Tasks:

Task	Hours	Definition of Done
False Cognates Highlighter	4	- Create false cognates database (20-30 pairs): - “assist” “assister” - “actually” “actuellement” - “attend” “attendre”- Warning badge appears on relevant questions- Tooltip explains difference- Add to item schema: <code>falseCognate: {en, fr, explanation}</code> - Works offline
Teacher Q&A Queue	6	- Student can ask question from practice screen- Question saved to IndexedDB: <code>{learnerId, itemId, question, timestamp}</code> - Teacher dashboard shows question queue- Teacher types answer, saves to IndexedDB- Next time student syncs, sees answer- UI: “ Ask Teacher” button- Works fully offline (store-and-forward)
User Testing Preparation	3	- Recruit 3-5 students (ideally Cameroonian or Francophone)- Prepare consent form (minimal risk, anonymous)- Prepare testing protocol: 1. Complete 1-2 modules 2. Try 3-5 new features 3. 10-min survey (SUS scale) 4. Optional 5-min interview- Setup screen recording (with permission)
User Testing Execution	6	- Conduct 3-5 sessions (remote or in-person)- Observe interactions, note issues- Collect survey responses- Record qualitative feedback- Document usability problems
Expert Review Coordination	4	- Send survey to 3-5 ESL educators- Survey questions: 1. Clarity of contrastive analysis (1-5 scale) 2. Cultural appropriateness of scenarios (1-5) 3. Usefulness of feedback (1-5) 4. Suitability for shared devices (1-5) 5. Likelihood to use in classroom (1-5) 6. Open feedback (text)- Follow up with 1-2 for brief interview- Collect responses
Evaluation Data Analysis	4	- Synthesize user testing findings- Calculate SUS score (if using SUS scale)- Identify top 3 usability issues- Analyze expert review responses- Write 2-3 page evaluation summary- Document findings for final paper
Advanced Analytics Features	5	- Item analysis: difficulty ranking (% correct)- Error pattern detection: most common wrong answers- Time-to-complete analysis per question- Comparative analytics: student vs baseline- Enhanced CSV exports with calculated metrics
Fixes Based on Feedback	4	- Prioritize top 3 usability issues- Implement fixes- Quick regression test- Document what changed and why

Total: 36 hours **Deliverable:** Evaluation data, advanced features, user-validated design

WEEK 16: Final Polish & Deliverables (Dec 9-15)

Theme: Documentation, presentation, deployment

Goals: - Landing page - Complete documentation - Final video - Academic paper - Deployment

Tasks:

Task	Hours	Definition of Done
Certificate Generator	3	- HTML Canvas generates certificate PNG- Shows: student name, module name, completion date, OffGrid English logo- Cameroonian colors (green/yellow/red border)- “Download Certificate” button- “Share Certificate” via Web Share API- Professional design (template-based)
Landing Page	6	- Create separate landing page: - Hero section (screenshot + tagline) - Features overview (6-8 key features) - Demo video embed (from Milestone 2) - Download/Install instructions - Testimonials (if available from expert review) - Footer: GitHub, contact- Responsive design- Deploy to Netlify/Vercel- SEO optimized (meta tags, Open Graph)
User Guide (PDF)	4	- Create 8-12 page PDF: - Getting started - How to practice - Understanding feedback - Using offline mode - Tracking progress - FAQ- Screenshots for each section- Simple, clear language- Available in app as downloadable PDF
Teacher Guide (PDF)	3	- Create 5-8 page PDF: - Using teacher validation - Interpreting analytics - Reviewing Q&A queue - Exporting data (CSV) - Best practices for classroom integration- Screenshots and examples
Technical Documentation	5	- Update README.md: - Architecture overview - Setup instructions - Build/deploy guide - API/data structure docs - Contributing guidelines- Architecture diagrams (update from proposal)- Privacy policy (no PII, anonymous ID)- License (MIT)- Changelog
Final Demo Video	6	- Record 10-min professional video: 1. Introduction (problem, solution) 2. Live demo (end-to-end user journey) 3. Highlight 10 key features 4. Show offline proof 5. Analytics demonstration 6. Teacher validation flow 7. CSV exports 8. Mobile responsiveness 9. Evaluation results summary 10. Future work- Professional editing (screen + voice)- Closed captions
Academic Paper (10-15 pages)	10	- Structure: 1. Introduction (problem, gap, vision) 2. Related Work (Assignments 1-3 synthesis) 3. Method (architecture, features, evaluation) 4. Results (evaluation findings, metrics) 5. Discussion (interpretation, limitations) 6. Future Work 7. Conclusion 8. References- Include figures: architecture, feature map, analytics screenshots- ACM or IEEE format
Final Presentation Slides	2	- 15-20 slides covering paper content- Visuals: screenshots, diagrams, data- Speaking notes
Deployment & Final Checks	3	- Deploy to Netlify (production)- Verify HTTPS works- Test PWA install on Android- Smoke test all features production- Verify analytics tracking (if added)- Final Lighthouse audit
GitHub Polish	2	- Add badges: PWA, TypeScript, License- Update README with live demo link- Add screenshots to README- Create GitHub releases/tags- Clean up branches- Archive development branches
Final Submission Preparation	2	- Create ZIP with: - Catalog.pdf (structure, build steps, Netlify URL) - Source code - Evaluation dossier - Sample CSVs - Playwright test results- Verify all files included- Test ZIP extraction

Total: 46 hours **Deliverable:** Complete portfolio-ready project with all documentation

CUMULATIVE HOURS BREAKDOWN

Week	Theme	Hours	Cumulative
11	Enhanced UI/UX + Analytics	28	28
12	Smart Learning Features	38	66
13	Mobile Excellence + Milestone 2	32	98
14	Testing & QA	28	126
15	Evaluation & Advanced Features	36	162
16	Final Polish & Deliverables	46	208

Total: ~208 hours (35 hrs/week average)

RISK MITIGATION

Risk	Probability	Impact	Mitigation
Voice Input browser compatibility	Medium	Medium	- Make optional with clear fallback- Test on 3 browsers- Graceful degradation documented
User testing recruitment (<3 participants)	Medium	Low	- Strengthen heuristic evaluation- Document as limitation- Use classmates as backup
Week 14-15 time crunch	Medium	Medium	- Tier 3 features can be reduced- Focus on Tier 1 (CORE) first- Certificate generator can move to “future work”
Evaluation data insufficient	Low	Medium	- Supplement with analytics from own testing- Use simulated learner data- Focus on feasibility over effectiveness
Bundle size exceeds 250KB	Low	High	- Remove unused Tailwind utilities- Tree-shake Recharts (import only needed charts)- Dynamic imports for routes- Use vite-bundle-visualizer to identify large deps
Mentor asks to reduce scope	Low	High	- Provide tiered fallback: - Plan A: All 21 features - Plan B: 10 CORE + 3 STRONG = 13 features - Plan C: 10 CORE only- All plans maintain offline-first + research grounding

SUCCESS CRITERIA

Technical Gates (Must Pass)

- PWA Lighthouse score 90
- Accessibility WCAG 2.1 AA compliant
- Playwright offline test: 0 network requests
- Bundle size 250KB gzipped
- Works on Android 8+ and iOS 14+
- TypeScript strict mode, 0 errors

Feature Gates (Must Complete)

- All 10 CORE features implemented and tested
- At least 3 STRONG features working
- Teacher validation + analytics functional
- CSV exports working for all data
- Offline mode proven (automated test)

Evaluation Gates (Must Document)

- User testing with 3 participants
- Expert review with 3 ESL educators
- Heuristic evaluation completed
- Evaluation summary (2-3 pages) written

Documentation Gates (Must Deliver)

- Academic paper (10-15 pages)
 - Final demo video (10 min)
 - User guide + Teacher guide
 - Technical documentation
 - Landing page deployed
-

WEEKLY CHECKPOINT QUESTIONS

Week 11

- Is the analytics dashboard with charts acceptable (vs tables-only proposal)?
- Are the Cameroonian scenarios culturally appropriate?
- Any concerns about dark mode implementation?

Week 12

- Is the personalized learning path algorithm too complex?
- Should spaced repetition use SM-2 or simpler intervals?
- Any feedback on achievement design?

Week 13

- Voice input: acceptable to make optional with fallback?
- Milestone 2 video submitted on time?
- Any scope adjustments needed?

Week 14

- Testing coverage sufficient?
- Any critical bugs blocking progress?

Week 15

- Evaluation data collection going well?
- Any ethical concerns with user testing?

Week 16

- Final deliverables aligned with expectations?
- Any last-minute adjustments needed?

TOOLS & LIBRARIES SUMMARY

Week 11 installations

```
npm install framer-motion recharts
```

Week 13 installations (if needed)

```
npm install react-swipeable
```

Week 14 installations

```
npm install -D @playwright/test @axe-core/react
```

Optional (Week 16)

```
npm install html2canvas jspdf # For certificate/PDF generation
```

Total Cost: \$0 (all MIT/Apache 2.0 licensed)

FALLBACK TIERS

Plan A: Full Implementation (21 features)

- All features as listed above
- Target: Week 16 completion

Plan B: Core + Strong (15 features)

- All 10 CORE features
- 5 STRONG features
- Defer: Challenge Mode, Certificate Generator, some Tier 3
- Target: Week 15 completion with buffer

Plan C: Core Only (10 features)

- 10 CORE features only
- All research-critical features maintained
- Document others as “Future Work”
- Target: Week 14 completion with extensive testing

Recommendation: Start with Plan A, drop to Plan B only if Week 13 checkpoint reveals delays.

NEXT STEPS

1. **Send mentor approval request** (with diagram + research grounding)
2. **Wait for mentor response** (expect 1-3 days)
3. **If approved:** Begin Week 11 tasks immediately
4. **If adjustments needed:** Revise plan and resubmit

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